Experiment 8: Insertion Sort with Duplicate Elements

Aim: To implement Insertion Sort on an array containing duplicate elements and analyze its time complexity.

Algorithm

1. Start the program.
2. Read the size of the array.
3. Read the array elements, including duplicate values.
4. Select the second element as the key.
5. Compare the key with the previous elements.
6. Shift all elements greater than the key one position to the right.
7. Insert the key into its correct position.
8. Repeat until all elements are sorted.
9. Display the sorted array.
10. Stop the program.

C

```
#include <stdio.h>

int main() {
    int a[100], n, i, j, key;
    printf("Enter the size of array: ");
    scanf("%d", &n);
    printf("Enter the array elements: ");
    for (i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }
    for (i = 1; i < n; i++) {
        key = a[i];
        j = i - 1;

        while (j >= 0 && a[j] > key) {
            a[j + 1] = a[j];
            j--;
        }
```

```
        a[j + 1] = key;
    }
    printf("Sorted array: ");
    for (i = 0; i < n; i++) {
        printf("%d ", a[i]);
    }
    return 0;
}
```

Input: 1 5 6 9 0 3 2 4 7 8

Output:

```
Enter the size of array: 10
Enter the array elements: 1 5 6 9 0 3 2 4 7 8
Sorted array: 0 1 2 3 4 5 6 7 8 9
```

Result:

Thus, the C program successfully sorts an array containing duplicate elements in ascending order using Insertion Sort and analyses its time and space complexity.